

Taylor Series Solutions of IVP's

$$y(x) = y(0) + \frac{y'(0)}{1!} (x-0) + \frac{y''(0)}{2!} (x-0)^2 + \frac{y'''(0)}{3!} (x-0)^3 + \dots + \dots$$

$$\text{So, } F(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$$

it would be more simpler to look at the practice problems we will do because this whole section is product rule and plugging into a equation

Practice Problems

Cannot be zero

#1) Find the first 3 non-zero terms of Taylor Polynomial given:

$$y' = x^2 + y^2, \quad y(0) = \boxed{1} \quad \text{1st term}$$

begin by plugging $y(0) = 1$ into y' :

$$y' = x^2 + y^2 \rightarrow (0)^2 + (1)^2 = 1 \Rightarrow y'(0) = \boxed{1} \quad \text{2nd term}$$

take the derivative of y' : think of this as explicit differentiation we do this because we have two y 's $y' = x^2 + y^2$

$$y'' = 2x + 2yy' \rightarrow 2(0) + 2(1)(1) \Rightarrow y''(0) = \boxed{2} \quad \text{3rd term}$$

We have gotten to the first 3 non zero terms we will now plug them into the equation on the top of the page

$$\boxed{1 + \frac{1}{1!}x + \frac{2}{2!}x^2}$$

#2) Find the first 4 non-zero terms of the Taylor Polynomial

$$y'' - 2xy' + x^2y = 0, \quad y(0) = 1, \quad y'(0) = -1$$

$\downarrow$

$y'' = 2xy' - x^2y$

↑
term 1 ↑
term 2

• this is zero it cannot be zero
remember we said none zero

$$y'' = 2xy' - x^2y \rightarrow 2(0)(-1) - (0)^2(1) \Rightarrow y''(0) = 0$$

you need to take the product rule

$f' \cdot g + f \cdot g'$

$$y''' = 2y' + 2xy'' - 2xy' - x^2y' \rightarrow 2(-1) + 2(0)(0) - 2(0)(1) - (0)^2(-1) \Rightarrow y'''(0) = -2$$

↑
term 3

$$y^{(4)} = 2y'' + 2y''' + 2xy'' - 2y' - 2xy' - x^2y''$$
$$\cancel{2(0)} + \cancel{2(0)} + \cancel{2(0)(-2)} - \cancel{2(1)} - \cancel{2(0)(-1)} - \cancel{2(0)(-1)} - (0)^2(0)$$
$$\Rightarrow y^{(4)}(0) = -2$$

↑
term 4

we now can just plug into the equation on the first page:

$$1 - \frac{1}{1!}x - \frac{2}{3!}x^3 - \frac{2}{4!}x^4$$

Find the first 6 non-zero terms of a Taylor series, centered at 0 when you are given:

$$y'' = x^2 + y^2 - 2y', \quad y(0) = 1, \quad y'(0) = 1$$

$\uparrow$ term 1 $\uparrow$ term 2

$$y'' = x^2 + y^2 - 2y' \rightarrow (0)^2 + (1)^2 - 2(1) \Rightarrow y''(0) = -1 \quad \text{term 3}$$

$$y''' = 2x + 2yy' - 2y'' \rightarrow 2(0) + 2(1)(1) - 2(-1) \Rightarrow y'''(0) = 4 \quad \text{term 4}$$

$$y^{(4)} = 2 + 2y'y' + 2yy'' - 2y''' \rightarrow 2 + 2(1)(1) + 2(1)(1) - 2(4) \Rightarrow y^{(4)}(0) = -6 \quad \text{term 5}$$

$$y^{(5)} = 2y''y' + 2y'y'' + 2y'y'' + 2yy''' - 2y^{(4)} \rightarrow 2(-1)(1) + 2(1)(-1) + 2(1)(-1) + 2(1)(4) - 2(-6)$$

$$y^{(5)}(0) = 14$$

$\uparrow$ term 6

$$y(x) = 1 + \frac{1}{1!}x - \frac{2}{2!}x^2 + \frac{4}{3!}x^3 - \frac{6}{4!}x^4 + \frac{14}{5!}x^5$$

For all of these we applied Product rule on them

#3) Find the first 6 non-zero terms of a Taylor series solution centered at zero:

$$y'' = x + y^2, \quad y(0) = 1, \quad y'(0) = 1$$

$\xrightarrow{\text{term 1}} \quad \xrightarrow{\text{term 2}}$

$$y'' = x + y^2 \rightarrow (0) + (1)^2 \Rightarrow y''(0) = 1$$

$\xrightarrow{\text{term 3}}$

$$y''' = 1 + 2yy' \rightarrow 1 + (1)(1) \Rightarrow y'''(0) = 3$$

$\xrightarrow{\text{term 4}}$

$$y^{(4)} = 2y'y' + 2yy'' \rightarrow 2(1)(1) + 2(1)(1) \Rightarrow y^{(4)}(0) = 4$$

$\xrightarrow{\text{term 5}}$

$$y^{(5)} = 2y''y' + 2y'y'' + 2y'y'' + 2yy''' \rightarrow 2(1)(1) + 2(1)(1) + 2(1)(1) + 2(1)(3) \Rightarrow y^{(5)}(0) = 12$$

$\xrightarrow{\text{term 6}}$

$$y(x) = 1 + \frac{1}{1!}x + \frac{1}{2!}x^2 + \frac{3}{3!}x^3 + \frac{4}{4!}x^4 + \frac{12}{5!}x^5$$

all of these we are required to take the product rule

$$2y'y' \quad F \cdot g$$

$$F' \cdot g + F \cdot g'$$

correct way $\rightarrow (2y'')(y') + (2y')(y'')$

when you are taking the derivative for F' do not do $2y' \rightarrow y'$ or $2y' \rightarrow y''$
remember you are treating these as explicit differentiation